CS341 · COMPUTER GRAPHICS

Midterm 451 Solutions

30 marks · 1 hour

Muharib

01 Solution Q1: short answers

1. Computer graphics is commonly understood to mean the **creation, storage, and manipulation of models and images**.

	Emissive (emitters)	Nonemissive (nonemitters)
Idea	Convert electrical energy into light	Use optical effects to convert sunlight or another light source into graphics patterns
Examples	Plasma panels · Thin-film electroluminescent · LED	LCD

02 Solution Q2: multiple choice

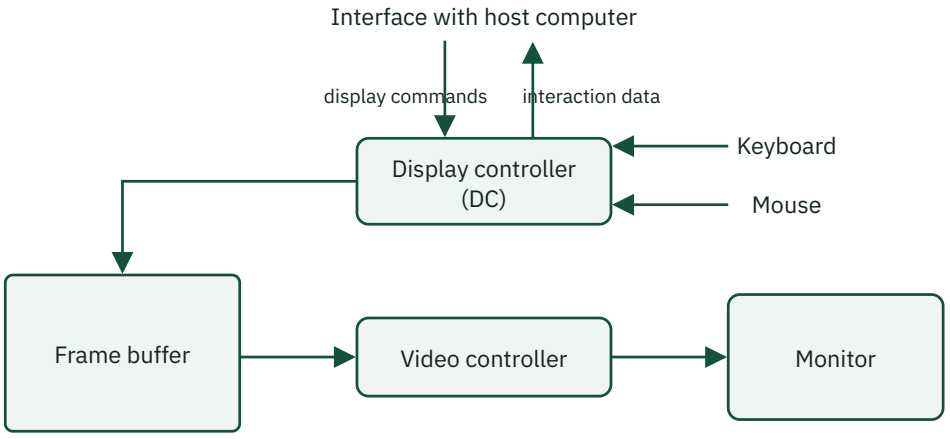
#	Answer	Reason
1	**b**	Raster stores the picture in the refresh (frame) buffer; the display list is for random scan.
2	**b**	The frame buffer is also called a bit map.
3	**a**	More pixels give a better image.
4	**d**	Refresh rate, not resolution.
5	**d**	
6	**d**	
7	**b**	
8	**b**	A screen point = pixel (picture element).
9	**d**	The CRT emits electrons; light appears when they hit the phosphor.
10	**b**	c is passive-matrix.

03 Solution Q3 and Q4

Operation	Result
Point – Point	**Vector**
Point + Vector	**Point**
Vector + Vector	**Vector**
Scalar * Vector	**Vector**
Scalar * Vector + Scalar * Vector	**Vector**

List	Answer
Hard-copy devices	Laser printer · Ink-jet printer · Film recorder · Electrostatic printer · Pen plotter
Video display devices	CRT (refresh CRT, DVST, color CRT) · LCD · Plasma panels · Thin-film electroluminescent displays

04 Solution Q5: the raster system



Slide 13 of Chapter 1. The doctor said there is no drawing this time, but the parts can still come as true or false.

05 Solution Q6: the same points with DDA

Bresenham and the circle are outside this term's scope; your problem is DDA. The same points solved with DDA for practice:

$$dx = 5 - 1 = 4 \qquad dy = 9 - 1 = 8$$

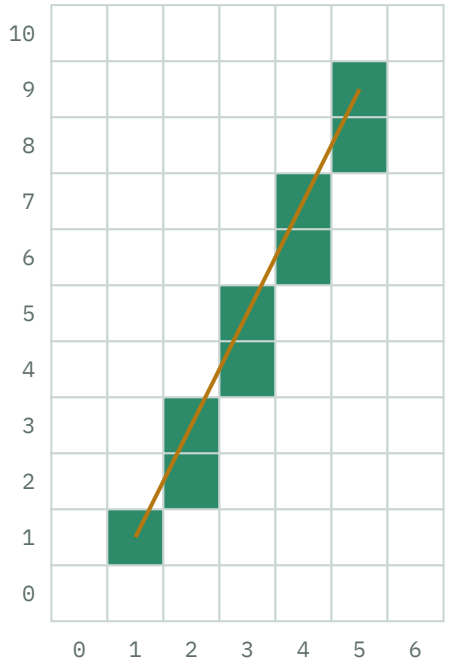
$$m = \frac{8}{4} = 2 \;\rightarrow\; |m| > 1 \;\rightarrow\; \text{step in y}$$

$$steps = \max(|dx|, |dy|) = 8$$

$$xInc = \frac{4}{8} = 0.5 \qquad yInc = \frac{8}{8} = 1$$

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round(a) = int(a + 0.5)
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k	x	y	round(x)	round(y)
0	1	1	1	1
1	1.5	2	2	2
2	2	3	2	3
3	2.5	4	3	4
4	3	5	3	5
5	3.5	6	4	6
6	4	7	4	7
7	4.5	8	5	8
8	5	9	5	9



Points: (1,1) (2,2) (2,3) (3,4) (3,5) (4,6) (4,7) (5,8) (5,9)